Editorial
Dealing With Technological Trajectories: Where We Have Come From and Where We Are Going

TECHNOLOGICAL progress is widely recognized as having considerable impact on human societies, which have become more and more dependent on human-made tools that entail intricate scientific processes, considered *technology* [item 1) in the Related Works]. The adoption of such tools is borne out of the necessity of survival at first [item 2) in the Related Works], and subsequently as a means to improve access to food, shelter, and clothing; arguably things that we cannot live without. Nevertheless, the urgency of understanding the role of technology in human life has never been as vital as today. Indeed, one could say that technological progress is the key ingredient driving human civilization: the invention of ever more complex tools traces the history of human civilization and its development. Each generation of technology lays the foundation for the next, the invention of simple machines, such as the wheel, for example, enabled the invention of the wheelbarrow, which enabled in turn building larger structures. In parallel, the wheel enabled the construction of gears to transmit power from crankshaft to driveshaft and also the rotation of magnetic tapes permitting the capture of binary data leading to the spinning hard disk drives and their superior digital data storage capacity.

Naturally, the role of technology as the fuel of societal change has not gone unnoticed and the social implications of technology emerged as the focus of a distinct field of study which flourished alongside the environmental and other new social movements in the 1960s. Domains of exploration initially considered the negative impacts that the First Industrial Revolution had on communities and the powerful capabilities afforded by the adoption of data-driven quantitative decision making during the Second World War. Breakthrough innovations, such as the creation of the atom bomb in the 1940s, which Jonathan D. Moreno called an "offset" [item 3) in the Related Works], one of several in the 20th century, was given special emphasis in this setting. Yet, the full implications of the technologies studied on human life required several decades to manifest and be evidenced, for example, the true costs of widespread nuclear energy use are perhaps becoming only visible today. Other technological innovations of the 19th and 20th centuries, such as the production of rubber and plastic, and the commensurate flow-on effects of deforestation and human-induced erosion in the form of soil salinity are setting contemporary global challenges, which are difficult to address due to the fact that the underlying processes are deeply ingrained in modern economies.

To be sure, the industrialization of production has resulted in extraordinary prosperity but also forced large rural populations seeking access to wealth to move to urban areas leading to previously unseen levels of migration. At the same time, the wealth gap between nation states has widened and so has the gap between better and less well-off areas even within the richest countries [item 4) in the Related Works]. Three billion of the world's population still cook over open wood fires [item 5) in the Related Works], and in many other parts of the world access to electricity is still scant or at least unstable. In 2018, UNESCO declared that "one in every five children, adolescents, and youth is out of school worldwide" [item 6) in the Related Works], and in 2016, they claimed this number was approximately 263 million children [item 7) in the Related Works]. According to the WHO, "globally, at least 2 billion people use a drinking water source contaminated with feces" [item 8) in the Related Works]. With 7.8 billion people in the world, that means more than 25% of the world is at risk of preventable death and disease, if only they had clean water to drink. The World Bank's preliminary forecast is that extreme poverty has declined to 8.6% in 2018, where poverty is defined as earning less than $1.90 a day [item 9) in the Related Works]. And yet we celebrate to be living through the Fourth Industrial Revolution [item 10) in the Related Works].

While still striving to distribute fairly the benefits of prosperity, new generations of technological innovation are occurring at an increasing pace. This, in turn, intensifies the limited opportunities to carefully consider their implications, directly impacting the world thus presenting often unexpected opportunities and challenges. Moreover, the complexity and dynamics of these new technologies makes their study harder. Convergence occurs at every level, and multidisciplinary approaches to understanding change, opportunities, and risks are critical. Helpful abstractions that were essential in making sense of earlier generations of technology, for example, the conceptual model established by Open Systems Interconnection for networked systems, fail to capture the full range of behaviors and capabilities emerging from the digital-physical convergence enabled by the Internet of Things and the deep interconnection of material entities and computational devices. We are dealing here with the emergence of complex and adaptive systems, and the interdependencies that we have been generally poor in mapping and measuring.

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We strive for flexibility, continual connectivity, detailed location information, and for complete knowledge [item 16) in the Related Works]. Forgetting the Socratic paradox *ipse se nihil scire id unum sciat* we often act as if we believe that we are in control of the future and the natural systems around us rather than one element of a complex network of interconnected systems. We have clearly set out sustainable development goals (SDGs) we wish to attain by 2030, placing our hopes in 17 areas that might secure our planet, and in the process our very selves [item 11) in the Related Works]. And while aims such as "1. No Poverty" and "6. Clean Water and Sanitation" seem to be obvious goals, we are dreaming of space travel for those very few of us who can afford it. What might be wrong with such innovation dreams given the aims of SDG "10. Reducing Inequality"? Might there be a clash between what we are imagining through our science-fiction dreams and what may be generally considered a human right to access to basic needs on Maslow's hierarchy [item 12) in the Related Works]? Is this simply a perpetual form of escapism, so we do not come face to face with our current and future states?

New technologies change society in profound ways in the long-term but also have concrete shorter-term repercussions in a changing geopolitical landscape. Notably, technological innovation influenced the relatively recent transformation of the cold world era super power balance by weakening their competitive agility. In more recent times, it has enabled East Asia countries to emerge as the largest consumer market globally, a development strategy further intensified by massive investments in artificial intelligence. In this multifaceted dynamic environment, nations make distinct choices with direct and wide-ranging societal ramifications often reflecting unbalanced tradeoffs, such as those between enhancing defense capabilities against addressing a more humane and equitable approach to the collective redistribution of food, shelter and opportunity. Furthermore, current technology choices trace distinct pathways between strengthening free speech and democracy as a principle of social organization against the control of every aspect of citizen behavior through state-of-the-art behavioral monitoring and predictive analytics, observing every detail of individual life reminiscent of Jeremy Bentham's Panopticon. Although the future may be uncertain, visions of the future diverge between those anticipating a technological utopia and the dystopian future predicted by social science fiction novels such as *Brave New World* and *1984*. Notwithstanding these antithetic views, how is it possible that we might break the poverty cycle and reach the stars at the same time? Are these conflicting aims, or one and the same? How could the "Tech for Good" vision put forward by stakeholders in academia, business, and government produce tangible results [item 13) in the Related Works]? Are considerations of the social impact of technology to be prioritized against financial and performance gains? What might "Responsible Research and Innovation" look like [item 14) in the Related Works], beyond considering "Ethical, Legal, Social Implications" in paradigms like ELSI that fostered basic and applied research with an anticipatory flavor initially in the context of the Human Genome Project (HGP)? How do we embed values in design? Should we encourage participation, through consultative and co-design-based approaches? And what about privacy and security by design, value-centered design and other ethics-based frameworks and methodologies?

And what of unanticipated and unintended consequences? For example, are our attempts to empower the citizens of developing countries with affordable solutions, such as one laptop per child (OLPC), or one mobile phone per person as has been the outcome in parts of Africa, South America, and Asia, achieving the intended SDG goals or are these a vehicle for negative externalities [item 15) in the Related Works]? And such are the issues befalling our newly industrialized communities propelled by ubiquitous access to the Internet, with potential triggers for known physical addictions and newly defined online addictions that peoples have previously had limited to no exposure. How might we forge forward with mindful behavior about what we want our future to be? What kinds of education might be necessary? Might some regulation be required or soft law? Which stakeholders should be involved in such decisions? These are all pertinent questions, and there are many more that demand exploration if we are to progress in a responsible manner.

In this IEEE TRANSACTIONS ON TECHNOLOGY AND SOCIETY, we are seeking papers that will help us to reimagine what the future might possibly be. In particular, we aim to foster a culture of thinking about technology and society that is not only guided by the economic and efficiency concerns but incorporates ethics and fairness as equal partners in decision making. We seek contributions that go beyond the utilitarian to envision how socio-technical transitions might become possible at scale; that ask how we can co-design for the future with the public interest integrated into the process of development; that explore how feedback loops from communities can be transformational in outcomes that matter to the local populations; and that question how new forms of governance and regulation can influence technological solutions to overcome the pacing problem. Furthermore, we aim to engage the research communities in discussions of how new workforces must be composed in a way to embrace diversity, gender equity and allow for flexibility in family contexts (e.g., for caring). How to develop new products and services in highly agile environments, providing access to retraining opportunities for those whose work has been displaced through automation. How we might address the question of global futures with g(local) solutions. The challenges are significant but so are the opportunities for new ways of doing and being.

In the inaugural issue, we have accepted four papers which typify the kind of research that is becoming increasingly important in socio-technical fields. These four papers are all regular papers, though it should be noted our *Transactions* also welcomes 6 page "short papers" and 2 page "brief papers." The assemblage of papers accepted in the first issue also may be considered examples for rigorous qualitative, quantitative, and experimental design-based methods for the transactions, among other methodological approaches.

The first paper by Krishnapriya *et al.* addresses facial recognition technology. The researchers from the Florida Institute of Technology and the University of Notre Dame challenge the controversy concerning possible bias due to accuracy varying based on race or skin tone. The team uses the ArcFace

matcher and the MORPH dataset in their experiments for the sake of reproducibility in the results, raising new questions that go beyond the technical aspects of their paper. While we are used to seeing public sector information (PSI) become open, for example, rights of way, electricity grids, and other infrastructure, the MORPH database includes high-resolution facial images "in full" of prison inmates over the time of their imprisonment. These are not unlike "freely available" U.S. Army images that are public data but the context is arguably inherently discriminatory from the outset. Most recently we have had major corporate investment into AI public-data based products, such as those offered by ClearView AI (https://clearview.ai/), and Aero Ranger (https://aeroranger.com/); products that are pushing the outer limits of ethical alignment and acceptability. These kinds of commercial products raise questions about power, the haves versus the have-nots, data as liquid gold, and the privatization of technologies that were once only in the hands of government and law enforcement authorities. One is left to ponder beyond the evidence provided by Krishnapriya *et al.* on the false match rates, and false nonmatch rates using a given algorithm, whether any matching is warranted under particular circumstances related to skin tone, race or any other attributes or characteristics.

The second paper is by Burr *et al.* and is a collaboration by researchers at the Alan Turing Institute and the Oxford Internet Institute. At its nascent stage "digital psychiatry," is addressed in this paper. The paper focuses on mental health disorders and the role that digital technologies can potentially play in improving mental health outcomes. The paper goes beyond simply using digital technologies for a patient to interact with their therapist, and looks at the development and use of artificial intelligence for assessing, diagnosing, and treating mental health issues outside traditional clinical settings. The pressure in varying sectors to know more about their workforce, their key constituents, their customers in education, employment, financial services, social media, and digital well being are explored. The paper identifies key problems and opportunities for public health, and offers recommendations for protecting and promoting public health and well being in information societies. The author team identify a real practical problem that requires a commensurate response.

The paper by Peters *et al.* is a great response to the issues raised in the preceding two papers. How do we navigate the potentialities of digital capabilities in this realm of new visibility, and in the name of care? On the one hand, we have Krishnapriya *et al.*, looking at facial recognition technology discrepancies using public data in individual matching that one could speculate might well be useful toward proving in the use of the technology for criminal cases, and on the other hand, we have Burr *et al.*, raising the practical issues of how digital technologies may well be used to determine one's mental health status. The team from Cambridge University, Imperial College London and the University of Sydney point to the traditional lack of process for attending to ethical impact within professional practice. They highlight the introduction of the P7000 series of standards within the IEEE as a means to address ethical issues in the design of autonomous and intelligent systems, with an emphasis on health technologies. The contribution of this paper is in describing two frameworks for integrating ethical analysis into engineering practice and the corresponding outcomes of an ethical analysis in the context of Internet-delivered therapy in digital mental health.

While the Peters *et al.*'s paper addresses the ethical requirement to consider unintended consequences of emerging technologies, Robertson *et al.*, from the University of Wollongong and Arizona State University, use mixed methods to investigate the dependencies of urban dwellers on fundamental technology systems that supply essential goods and services. Robertson *et al.*, claim we can reduce end-user vulnerability by identifying the feasibility of less exposed technological systems, using a "theory of exposure." It is argued that by knowing what the cause of exposure might be, we can address these in the initial designs of new technologies, but more importantly, provide levels of redundancy in existing systems that are less prone to typical failures experienced by society that impact large numbers of people. Robertson *et al.* focus on technological issues and also raise questions about data privacy, citizen rights, and the "right to repair."

A consistent theme across all four papers is one of sensitive data: our ethical use of this data for secondary purposes, a reduction in exposure, and a minimization of anticipated negative unintended consequences. The papers collectively allude to an underlying notion presented in this editorial; that is, just because something is technically feasible, it should not necessarily be immediately implemented. Rather, we must move forward with empathy by investigating design options and socio-ethical implications in a manner that strives to achieve minimal harm to individual citizens, customers, and society at large.

Finally, a special mention to Prof. John Impagliazzo of Hofstra University who is also the VP of Publications at the IEEE Society on the Social Implications of Technology (IEEE SSIT) for his ongoing dedication to seeing this Transactions take life. This project was conceived in the beginning of 2017, and without his commitment it just would not be. We thank, especially, our Administrator Mr. Salah Hamdoun (Ph.D. Candidate, International Global Development Program, School for the Future of Innovation in Society at Arizona State University) who has worked tirelessly with the incoming manuscripts we are receiving and for liaising with Co-Editors and Associate Editors for timely reviews. We are aiming to get feedback to authors within as short a time frame as possible without compromising on quality and depth of feedback. We thank Sonal Parikh of IEEE for her Manuscript Central support and her dedication to making the online transaction portal functional for our prospective authors. And to our Managing Editor AndreAnna McLean and Katie Sullivan, Manager, Journals Production thank you for your enthusiasm and precision. We would also like to thank the Board of Governors of IEEE SSIT who supported the initiative to go forward with what looks like will be a very popular publication for interdisciplinary researchers and multidisciplinary engineers. We encourage authors from all over the world, to submit their research to us, reviewing the scope of the transactions which is noted here in full. We publish "research papers on the interactions among technology, science, and society; on the impact of such interactions on individuals and society; and on the ethical, professional and

social responsibility in the practice of science, technology, engineering, and mathematics. The publication also provides a forum for open discussion of resulting issues."

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